

DEPARTMENT OF CSE, CS&IT, AI&DS
COURSE CODE: 24SDCS02 / 24SDCS02E / 24SDCS02P / 24SDCS02L
FULL STACK APPLICATION DEVELOPMENT

Date of the Session: __/__/____

Time of The Session: ____to____

#SKILL - 11 → React API Integration - Fetching Data Using Fetch API, Axios & Local JSON**Prerequisites:**

- **Basic understanding of React**
- **Knowledge of useEffect and useState hooks**
- **Understanding of API calls (fetch / axios)**

A news portal wants to fetch and display user and article information dynamically using **React**. Developers must learn how **useState**, **useEffect**, **fetch()**, **axios**, and **local JSON** work. This Skill demonstrates how frontend applications interact with **local data**, **public APIs**, and **fake APIs**, and how different components are navigated using hyperlinks.

Tasks:**Part A – Fetching from Local JSON**

1. Create a **users.json** file inside the public folder with 5–6 user records, and build a **LocalUserList** component using **fetch()** to load and display the data with name, email, phone, and loading/error states.

Part B – Fetching from JSONPlaceholder

2. Create a **UserList** component using **fetch()** to call <https://jsonplaceholder.typicode.com/users>
Display name, email, phone with loading and error handling.

Part C – Fetching from Fake API

3. Create a **FakePostList** component using **axios** to call **any ONE API**, such as:
 - a. <https://dummyjson.com/posts> OR
 - b. <https://fakestoreapi.com/products>

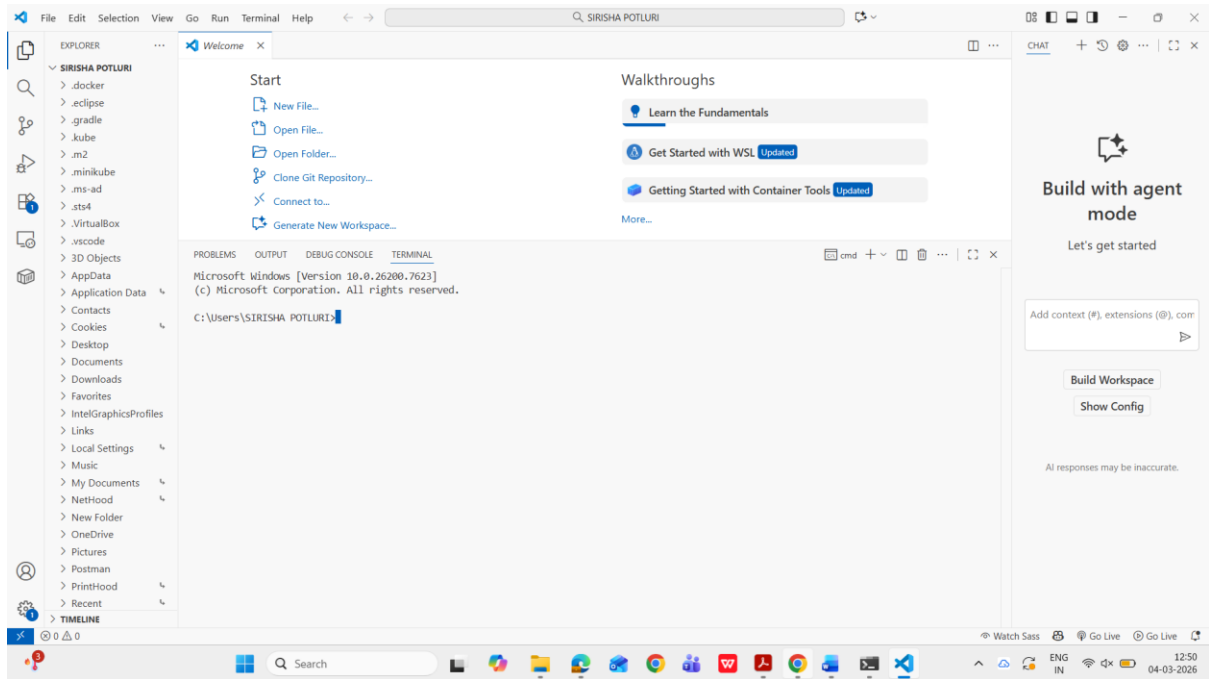
Display fields such as title, body, or description, and add a **Refresh** button to reload data.

Part D – Integration & Navigation

4. Create a **Dashboard/Home component** containing three hyperlinks/buttons:
 - a. Local Users
 - b. Users API
 - c. Fake API Posts

Each link should open its respective component.

5. Import **Dashboard** into **App.jsx** so all components can be accessed through hyperlinks instead of writing them directly in App.js.
6. Add a **dropdown filter** in FakePostList (or PostList) to filter data based on API fields (userId or category).
7. Apply **basic CSS styling** to keep the UI neat and readable.
8. **Push the React project to GitHub.**



Project Setup

Create Project

```
npm create vite@latest fsad-lab
```

```
cd fsad-lab
```

```
npm install
```

```
npm install axios react-router-dom
```

Folder Structure

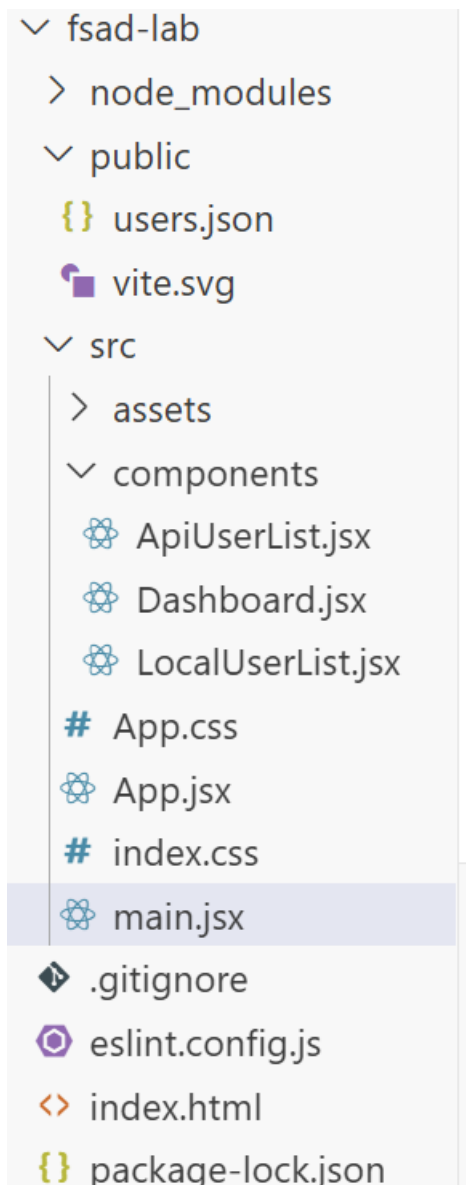
```
src/
```

```
|— components/  
|   |— Dashboard.jsx  
|   |— LocalUserList.jsx  
|   |— ApiUserList.jsx
```

```
|— App.jsx  
|— main.jsx
```

```
public/
```

```
|— users.json
```



Local JSON (users.json)

Create File

Add inside public/users.json

```
[
  {
    "id": 1,
    "name": "John Doe",
    "email": "john@gmail.com",
    "phone": "1234567890"
  },
  {
    "id": 2,
```

```

    "name": "Alice Smith",
    "email": "alice@gmail.com",
    "phone": "9876543210"
  },
  {
    "id": 3,
    "name": "David Lee",
    "email": "david@gmail.com",
    "phone": "5555555555"
  }
]

```

LocalUserList.jsx

```

import { useEffect, useState } from "react";

function LocalUserList() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

  useEffect(() => {
    fetch("/users.json")
      .then(res => res.json())
      .then(data => {
        setUsers(data);
        setLoading(false);
      })
      .catch(() => {
        setError("Error loading data");
        setLoading(false);
      });
  }, []);

  if (loading) return <h3>Loading...</h3>;
  if (error) return <h3>{error}</h3>;

  return (
    <div>
      <h2>Local Users</h2>
      {users.map(user => (
        <div key={user.id}>
          <p>{user.name}</p>
          <p>{user.email}</p>
          <p>{user.phone}</p>
          <hr />
        </div>
      ))}
    </div>
  );
}

```

```

    ))}
  </div>
);
}

export default LocalUserList;

```

ApiUserList.jsx

```

import { useEffect, useState } from "react";

function ApiUserList() {
  const [users, setUsers] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState("");

  useEffect(() => {
    fetch("https://jsonplaceholder.typicode.com/users")
      .then(res => res.json())
      .then(data => {
        setUsers(data);
        setLoading(false);
      })
      .catch(() => {
        setError("Error fetching API");
        setLoading(false);
      });
  }, []);

  if (loading) return <h3>Loading...</h3>;
  if (error) return <h3>{error}</h3>;

  return (
    <div>
      <h2>API Users</h2>
      {users.map(user => (
        <div key={user.id}>
          <p>{user.name}</p>
          <p>{user.email}</p>
          <p>{user.phone}</p>
          <hr />
        </div>
      ))}
    </div>
  );
}

```

```
export default ApiUserList;
```

Dashboard.jsx

```
import { Link } from "react-router-dom";

function Dashboard() {
  return (
    <div>
      <h1>Dashboard</h1>

      <ul>
        <li><Link to="/local">Local Users</Link></li>
        <li><Link to="/api">Users API</Link></li>
      </ul>
    </div>
  );
}

export default Dashboard;
```

App.jsx (Routing)

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import Dashboard from "../components/Dashboard";
import LocalUserList from "../components/LocalUserList";
import ApiUserList from "../components/ApiUserList";

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Dashboard />} />
        <Route path="/local" element={<LocalUserList />} />
        <Route path="/api" element={<ApiUserList />} />

      </Routes>
    </BrowserRouter>
  );
}
```

```
);  
}
```

```
export default App;
```

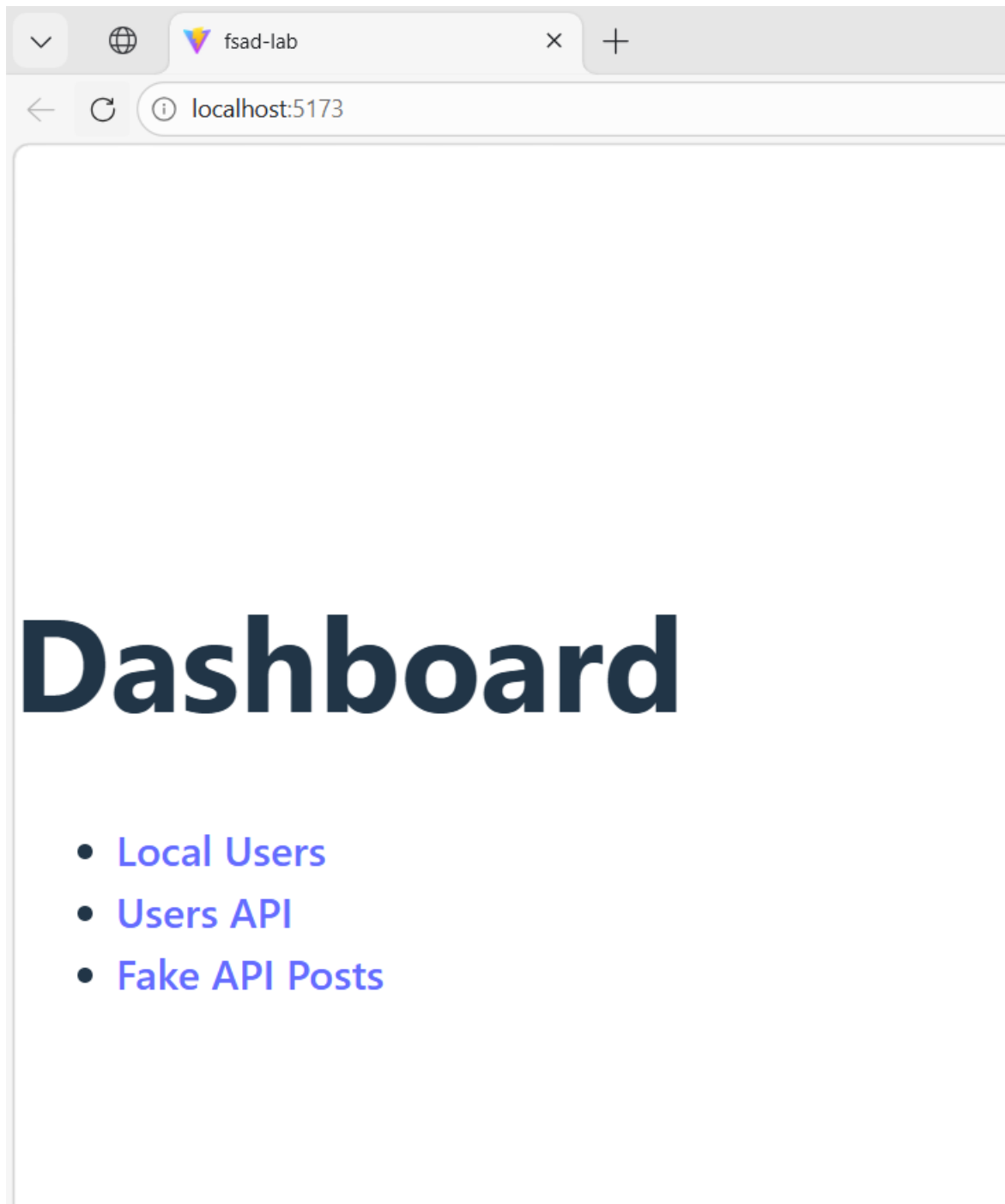
main.jsx

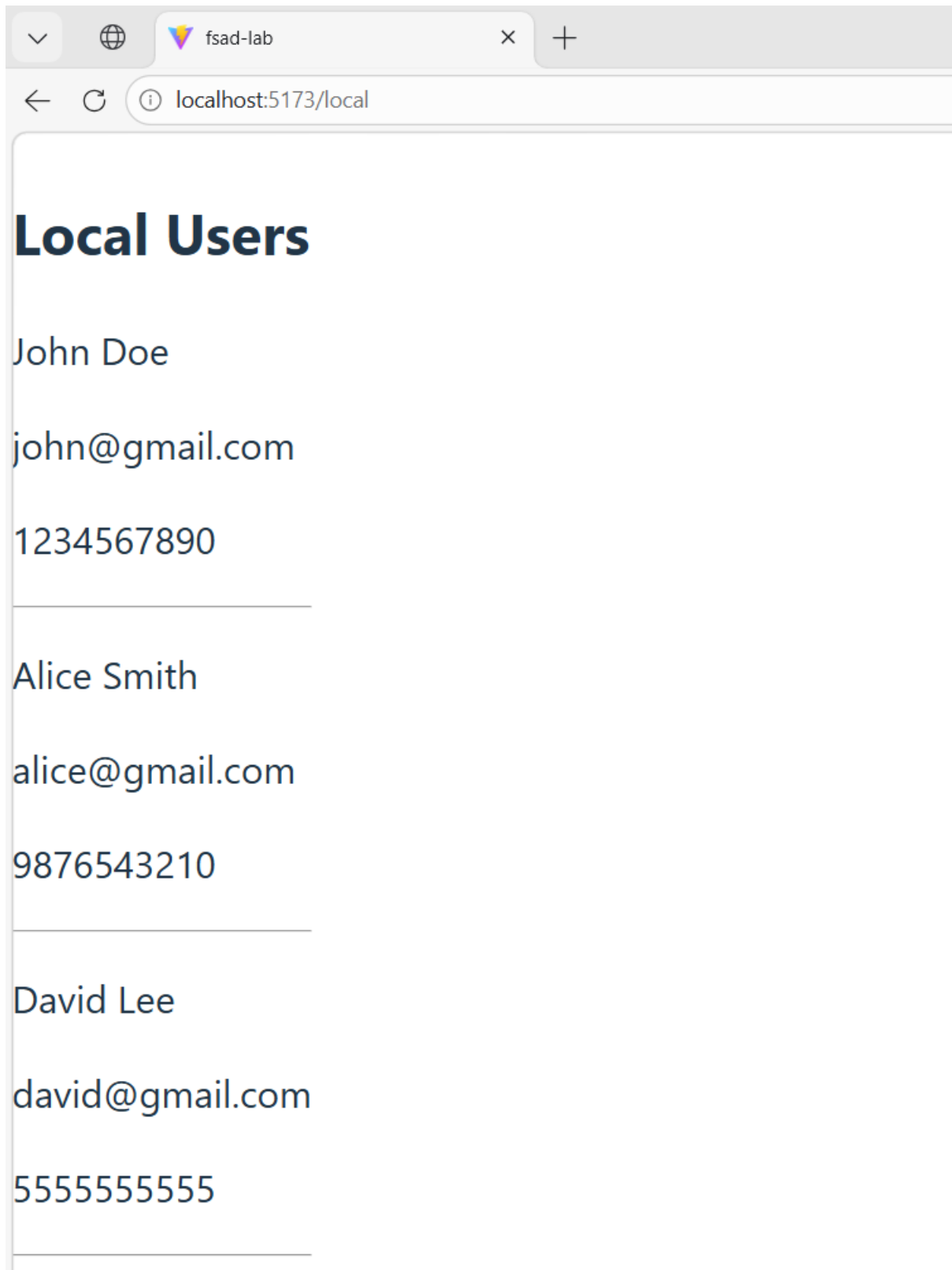
```
import { StrictMode } from 'react'  
import { createRoot } from 'react-dom/client'  
import './index.css'  
import App from './App.jsx'  
  
createRoot(document.getElementById('root')).render(  
  <StrictMode>  
    <App />  
  </StrictMode>,  
)
```

Run Project

```
npm run dev
```

```
→ Local:   http://localhost:5173/  
→ Network: use --host to expose  
→ press h + enter to show help
```





Local Users

John Doe

john@gmail.com

1234567890

Alice Smith

alice@gmail.com

9876543210

David Lee

david@gmail.com

5555555555

fsad-lab

×

+

←

↻

localhost:5173/api

API Users

Leanne Graham

Sincere@april.biz

1-770-736-8031 x56442

Ervin Howell

Shanna@melissa.tv

010-692-6593 x09125

Clementine Bauch

Nathan@yesenia.net

1-463-123-4447
